

Sedat Sekizkardes

Data mining 01

Market Basket Analysis Using Association Rule Mining

1. Introduction

Data mining is the process of extracting meaningful patterns and insights from large datasets. In the retail industry, data mining plays a critical role in understanding customer purchasing behavior and improving business strategies. One of the most widely used techniques in retail data mining is market basket analysis, which identifies relationships between products that are frequently purchased together.

This project focuses on analyzing a transactional retail dataset to discover associations between items using association rule mining. By identifying these relationships, businesses can improve product recommendations, optimize store layouts, and increase sales through effective cross-selling strategies. For example, online platforms such as Amazon use similar techniques to suggest products with the “Customers who bought this also bought” feature.

The main objectives of this project are:

- To identify frequently purchased items
- To discover strong associations between products
- To interpret these associations in a real-world retail context

To achieve these goals, exploratory data analysis (EDA) and the Apriori algorithm were applied to the dataset.

2. Methods

2.1 Data Preprocessing

Before applying any data mining techniques, the dataset was carefully cleaned and prepared. Data preprocessing is a crucial step because the quality of the input data directly affects the quality of the results.

The following steps were performed:

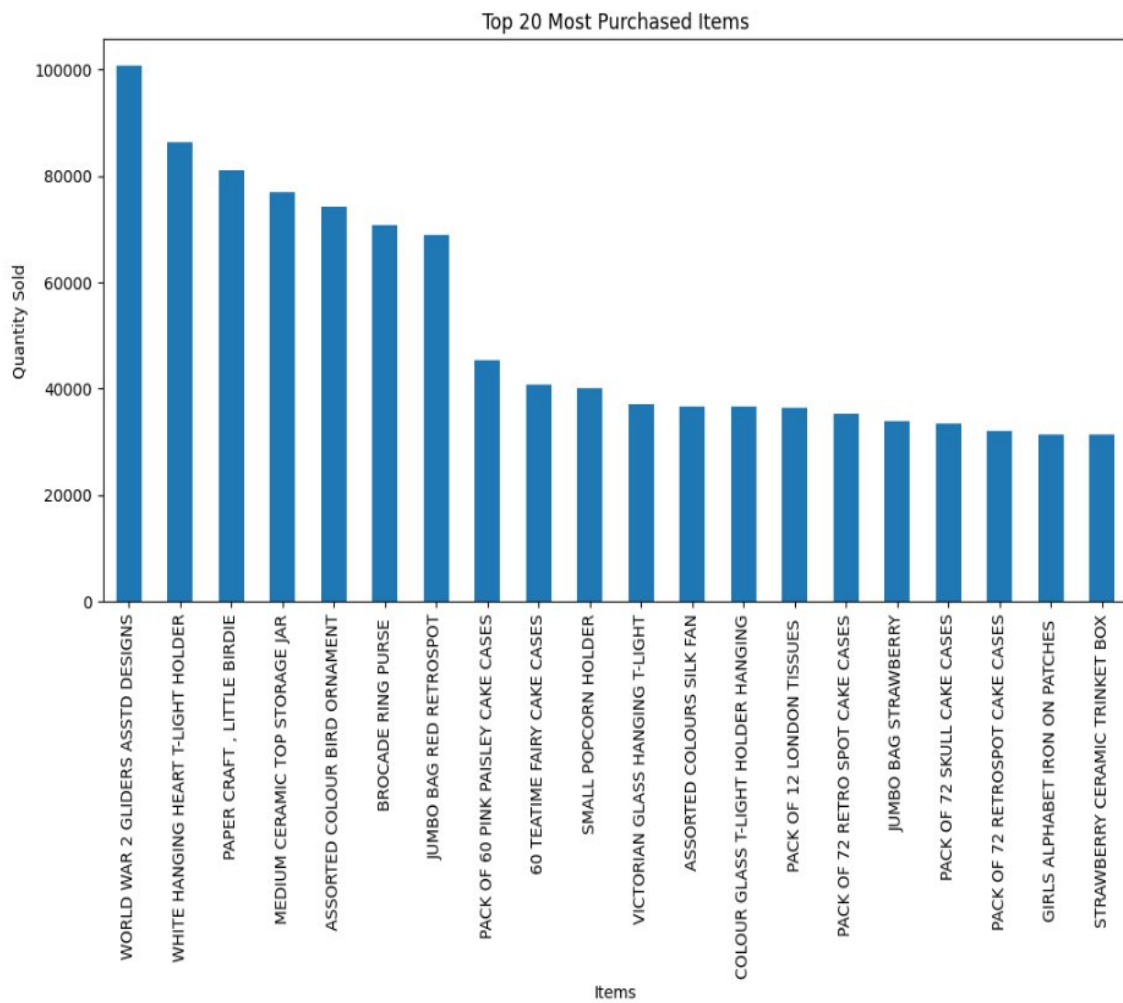
- Removed missing values to avoid incomplete transactions
- Filtered out canceled or invalid transactions
- Selected transactions from the United Kingdom to ensure consistency in purchasing behavior
- Limited the dataset to the top 200 most frequent items to reduce computational complexity and focus on meaningful patterns

The decision to focus on UK transactions introduces a limitation in terms of generalizability, but it allows for a more consistent and interpretable analysis. Similarly, restricting the dataset to the most frequent items helps improve performance and reduces noise in the data.

After cleaning, the dataset was transformed into a transactional format, where each invoice represents a list of purchased items. This format is required for association rule mining.

2.2 Exploratory Data Analysis (EDA)

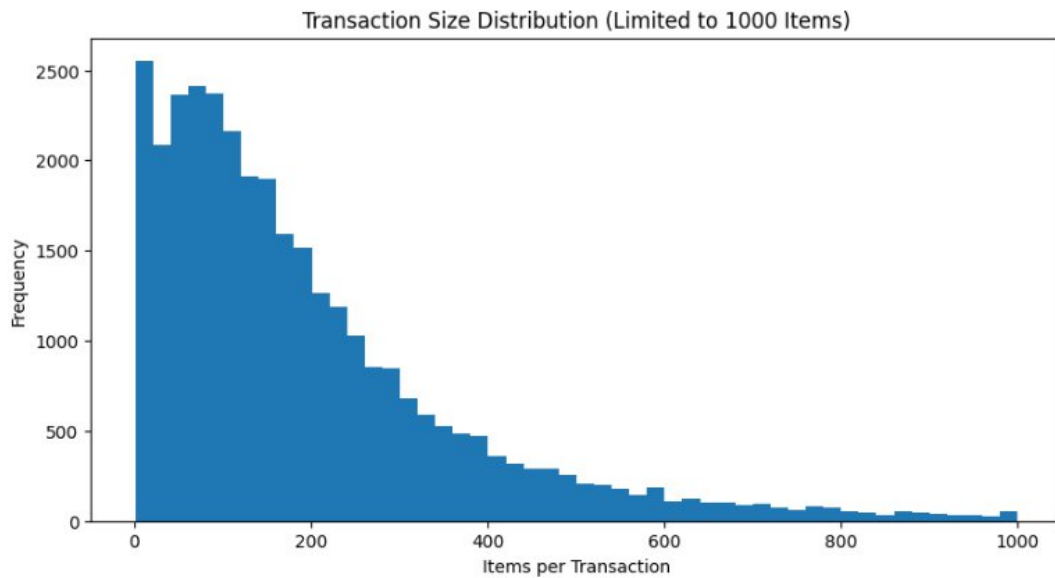
Exploratory Data Analysis was conducted to better understand the dataset and identify trends before applying the Apriori algorithm.



This graph highlights the most frequently purchased items in the dataset.

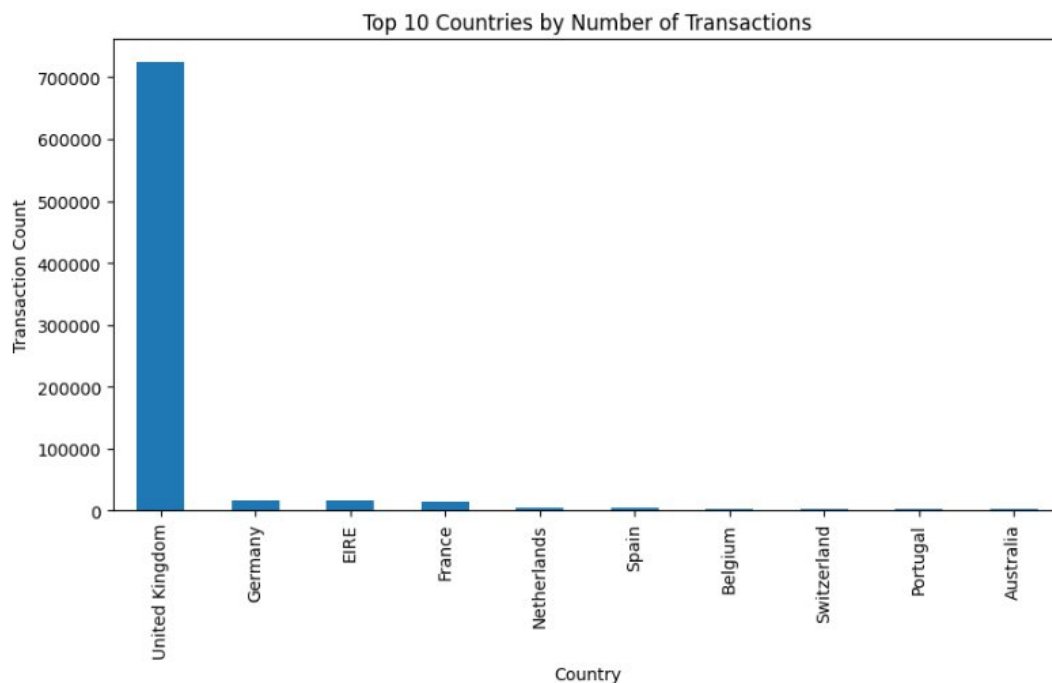
The presence of highly dominant products suggests that certain items appear in a large number of transactions. This is important for association rule mining because frequent items are more likely to form strong rules with other products.

As a result, these items play a key role in discovering meaningful patterns using the Apriori algorithm.



This graph illustrates the distribution of transaction sizes. Most transactions contain a small number of items, while only a few transactions include a large number of products.

This indicates that customers typically purchase a limited number of items per visit. For the Apriori algorithm, this means that most frequent itemsets will be small, and large item combinations will be relatively rare.



This graph shows the distribution of transactions across countries. The United Kingdom clearly dominates the dataset, with a significantly higher number of transactions

compared to other countries. This justifies focusing the analysis on UK transactions to maintain consistency and reduce noise. However, this decision also limits the generalizability of the results to other regions, which is an important limitation of the study.

2.3 Association Rule Mining (Apriori Algorithm)

The Apriori algorithm was used to identify frequent itemsets and generate association rules. It works by iteratively identifying combinations of items that appear frequently together in transactions.

The algorithm relies on three key metrics:

- Support: The proportion of transactions that contain a particular itemset
- Confidence: The likelihood that a consequent item is purchased given the antecedent item
- Lift: A measure of how much more likely items are purchased together compared to random chance

Lift is particularly important because it indicates the strength of a relationship beyond coincidence. A lift value greater than 1 suggests a positive association, while higher values indicate stronger relationships.

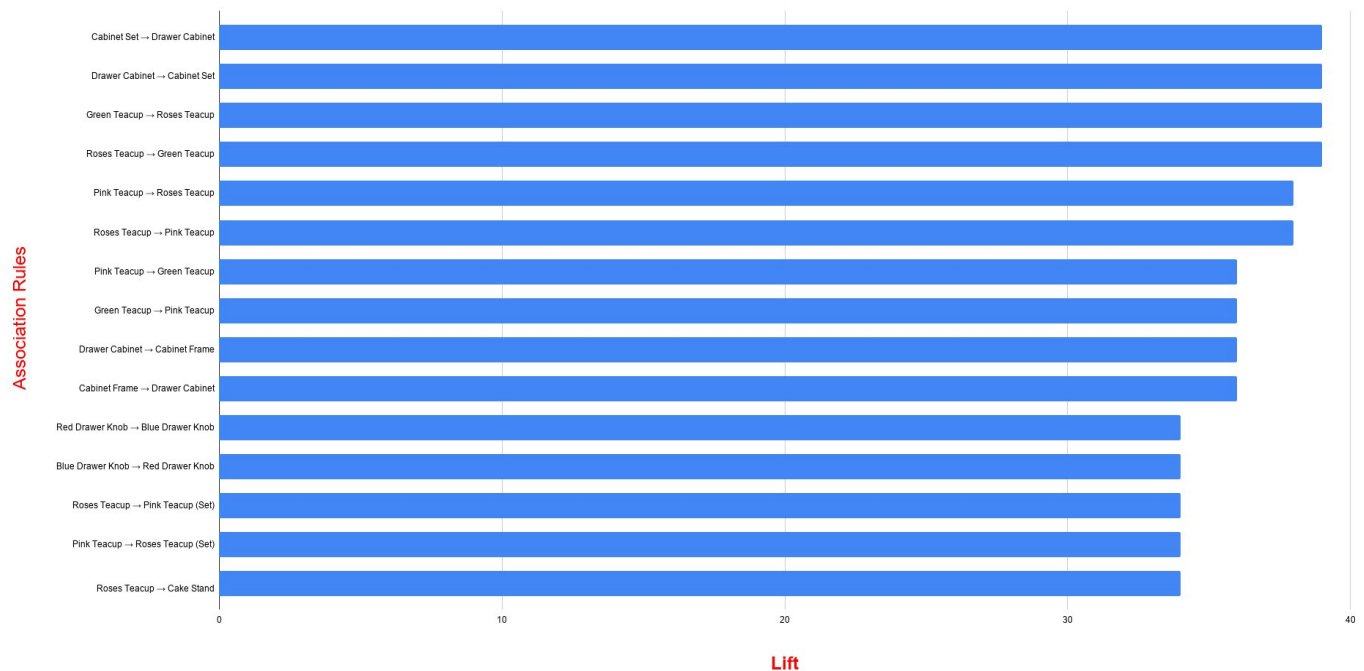
The following parameters were used:

- Minimum support: 0.02
- Minimum confidence: Not directly specified; rules were filtered using a minimum lift threshold of 1.0 to identify meaningful associations.
- Lift used as the primary ranking metric

3. Results

The Apriori algorithm generated multiple association rules, from which the strongest ones were selected based on lift values.

Top Association Rules by Lift



The chart displays the top association rules ranked by lift, which measures how strongly two products are related compared to random chance. Higher lift values indicate stronger and more meaningful relationships between items.

Several important patterns can be observed from these results:

- Cabinet-related items frequently appear together, suggesting that customers tend to purchase furniture components as part of a set rather than individually.
- Teacup and saucer combinations show strong associations, which is expected due to their complementary functional relationship. This confirms that the Apriori algorithm successfully captures logical product pairings.
- Decorative and themed items are often purchased together, indicating that customers follow aesthetic or thematic preferences when shopping.

Overall, these patterns demonstrate that customers tend to purchase related or complementary products together. These insights can be valuable for retail strategies such as product placement, recommendation systems, and bundled promotions.

4. Interpretation

The results provide meaningful insights into customer purchasing behavior. High lift values indicate strong associations that go beyond random coincidence.

For example:

- The rule “Cabinet Set → Drawer Cabinet” suggests that customers purchasing cabinet sets are highly likely to also purchase drawer cabinets
- Teacup-related rules indicate that customers often buy matching or complementary tableware items
- Decorative items appearing together suggest that customers are purchasing items based on style or theme

These findings have several practical applications:

- Recommendation systems: Suggesting related products to customers
- Store layout optimization: Placing related items near each other
- Marketing strategies: Creating bundled offers or promotions

5. Critical Assessment

While the results are useful, several limitations must be considered.

5.1 Validity

The analysis is based on a subset of the data (UK transactions and top 200 items), which may limit the generalizability of the results. Different regions may exhibit different purchasing behaviors.

5.2 Limitations

- The dataset may contain seasonal patterns that are not accounted for
- Limiting to top items excludes less frequent but potentially important products
- Association rules identify correlations but do not imply causation
- The Apriori algorithm can be computationally expensive for larger datasets

5.3 Ethical Considerations

Using customer transaction data raises privacy concerns. It is important to ensure that:

- Data is anonymized
- Customer information is protected
- Insights are used responsibly

Businesses must balance data-driven decision-making with ethical data usage.

6. Conclusion

This project successfully applied association rule mining to identify patterns in retail transaction data. The Apriori algorithm revealed strong relationships between products, particularly among complementary items.

The findings demonstrate how data mining techniques can provide valuable insights for business decision-making. By leveraging these insights, retailers can improve customer experience and increase sales.

Future work could include:

- Applying clustering techniques for customer segmentation
- Expanding the dataset to include multiple regions
- Using more advanced algorithms for improved performance